INTERIM REPORT

Cancer Patient Care System



A Digital Cancer Patient Care System Connecting Patients, Doctors, Donors, and Hospital Services

Prepared by **IS - 16**
Second-Year Group Project

University of Colombo School of Computing (UCSC)

1. Introduction 4

1.1 Domain Description 4

1.2 Current System and Limitations 4

1.3 Goal and Objectives 5

1.4 Assumptions 5

2. Feasibility Study 6

2.1 Technical Feasibility 6

2.2 Economic Feasibility 6

2.3 Schedule Feasibility 6

2.4 Operational Feasibility 6

2.5 Legal and Ethical Feasibility 7

3. Requirements 8

3.1 Stakeholders / Actors 8

3.2 Functional Requirements 8

Patient 8

Doctor 8

Medical Staff 9

Pharmacist 9

Administrator 9

3.3 Non-Functional / Quality Attribute Requirements 10

3.4 In Scope 10

3.5 Out of Scope 11

3.6 Constraints and Limitations 11

3. Proposed System Architecture 12

4.1 Architecture Overview 12

4.2 Components and Functionalities 14

4.3 Component Interactions 14

4. System Design Diagrams 15

5.1 Use Case Diagram 15

5.2 Class Diagram (Proposed Interim Design) 16

5.3 ER Diagram (Proposed Interim Design) 18

5.4 Activity Diagram (Example: Appointment Booking) 19

6. Current Progress 20

6.1 Remaining Work 20

6.2 Member Contributions / Responsibilities 21

7. Technologies 22

8. Conclusion 23

Name of the Supervisor	**Signature of the Supervisor**	**Date**
Prof. M.G. Noel A.S. Fernando (PhD)		2026.09.23
Name of the Co-Supervisor	**Signature of the Co-Supervisor**	**Date**
Ms. M.A.S.U. Fernando		2026.09.23
Name of the Client	**Email Address**	
Prof. M.G. Noel A.S. Fernando (PhD)	nas@ucsc.cmb.ac.lk	

1. Introduction

1.1 Domain Description

The Cancer Patient Care System is a centralized web-based healthcare information system designed to support cancer-patient services and hospital operations. It connects six main actors: Patient, Doctor, Medical Staff, Pharmacist, Benefactor , and Administrator. The system brings together patient registration, appointments, medical records, prescriptions, medicine availability, Benefactor support, transport schedules, nutritional guidance, motivational resources, and administrative reporting.

The project is motivated by the need to reduce disconnected and paper-based activities in cancer-care environments and to improve coordination among patients and hospital stakeholders.

1.2 Current System and Limitations

The current environment relies heavily on manual or disconnected processes. This creates delays, duplicated work, difficulty accessing centralized information, and communication gaps among hospital departments.

- Patients may experience long waiting times for registration and appointments and may have difficulty accessing medical records, prescriptions, treatment information, and support services.
- Doctors and medical staff must handle large volumes of records and time-consuming documentation while coordinating appointments and treatment schedules.
- Benefactor information and donation activities may be handled separately, reducing visibility and coordination.
- Administrators may have difficulty producing timely reports because information is fragmented across processes.

1.3 Goal and Objectives

The main goal is to design and develop a secure, reliable, user-friendly web-based Cancer Patient Care System that improves hospital efficiency and the overall patient-care experience.

- Digitalize patient registration and profile handling.
- Store and access patient medical records securely.
- Improve appointment booking, approval, rescheduling, and tracking.
- Support digital prescriptions and treatment-progress updates.
- Track medicine availability in pharmacy.
- Support Benefactor registration, donation information, and donation history.
- Provide transport schedules, meal plans, and motivational resources to patients.
- Generate operational and analytical reports for administrators.

1.4 Assumptions

- Patients provide accurate and up-to-date personal and medical information.
- Doctors, medical staff, pharmacists, Benefactors, and administrators use the system according to assigned roles.
- The database server and hosting environment remain available during system use.
- Users have suitable devices and internet access.
- Required development tools and project resources remain available during the project period.

2. Feasibility Study

2.1 Technical Feasibility

The system uses technologies already selected for the project: HTML, CSS, and JavaScript for the frontend; PHP for server-side processing; MySQL for data storage; and Apache through XAMPP for local development. Visual Studio Code and GitHub support development and version control. The project intentionally avoids frameworks and external libraries, keeping the implementation aligned with the team's academic constraints.

2.2 Economic Feasibility

Development is carried out by undergraduate students as part of coursework. The main software tools are free or open source, development and testing can be performed locally using XAMPP, and the team uses existing laptops and internet connections. Therefore, the direct financial cost during development is minimal.

2.3 Schedule Feasibility

The proposal plans the project across approximately one academic year. Requirements and design are followed by frontend, backend, database implementation, integration, testing, documentation, presentation preparation, and final submission. Regular supervisor feedback is used to refine the work throughout development.

2.4 Operational Feasibility

The proposed system is browser-based and uses role-based dashboards. Patients can access appointments, medical information, prescriptions, support resources, and transport information, while hospital users perform role-specific tasks. Centralized information is expected to reduce paperwork and improve coordination.

2.5 Legal and Ethical Feasibility

The system handles sensitive medical information. Patient privacy, confidentiality, consent, secure authentication, role-based authorization, and controlled access to medical records are therefore essential design considerations.

3. Requirements

3.1 Stakeholders / Actors

- Patient
- Doctor
- Medical Staff
- Pharmacist
- Benefactor
- Administrator

3.2 Functional Requirements

Patient

- Securely log in/log out.
- View and update personal profile information.
- View appointment status/schedules.
- View medical reports and treatment history.
- View prescribed medications.
- View transport schedules.
- Access meal plans and nutrition guidance.
- Access awareness and motivational resources.

Doctor

- Create and update patient medical status and treatment details.
- Create digital prescriptions.
- Review patient treatment progress.
- Add meal plans for the patients
- Suggest appropriate movitational resources

Medical Staff

- Patient Registration
- Update patient information.
- Coordinate appointment schedules and patient queues.
- Show Doctor Availability.
- Upload and maintain medical reports.
- Managing patient support requests
- Coordinating with the benefactors and confirming approved assistance

Pharmacist

- View available medicines and current stock quantities.
- Add medicines and update stock quantities.

Benefactor

- Register and maintain Benefactor profiles.
- Submit donation information.
- View previous donation history.

Administrator

- Create, view, update, and delete user accounts and permissions.
- Monitor system activities.
- Oversee benefactor activities and contributions.
- Generate operational and analytical reports.
- Update system settings and configurations.

3.3 Non-Functional / Quality Attribute Requirements

Quality Attribute	Requirement
Reliability	Use validation, exception handling and error logging to reduce failures and incorrect data.
Performance	Use efficient database queries, indexing and server-side optimization for acceptable response times.
Availability	Use a reliable hosting environment with regular maintenance and monitoring.
Usability	Provide a clear, consistent and responsive interface that is easy for patients and hospital staff to learn.
Scalability	Use a database and modular structure that can support future enhancements and more users.
Maintainability	Apply standard coding practices and clear modular organization.

3.4 In Scope

- Patient registration, profile updates, medical-history storage and retrieval.
- Appointment booking, status tracking, approval/rejection/rescheduling and schedule coordination.
- Doctor access to records, medical-status updates, treatment monitoring and digital prescriptions.
- Medicine availability and stock-quantity updates.
- Benefactor registration, donation information and donation history.
- Meal plans, motivational resources and transport-schedule viewing.
- User administration, system monitoring and report generation.

3.5 Out of Scope

- Online payments or financial billing.
- AI-based disease diagnosis or prediction.
- Full hospital HR management.
- Integration with external national health databases.
- Live telemedicine video consultations.
- Full medicine inventory management beyond availability/quantity tracking.
- Online pharmacy registration.

3.6 Constraints and Limitations

- Medical records are restricted to authorized users.
- A stable internet/network connection is required for online access.
- Appointments depend on doctor availability and hospital operating hours.
- Medicine data depends on timely pharmacist updates.
- The system must be completed within the academic project timeline.
- Performance depends on available hardware and network infrastructure.

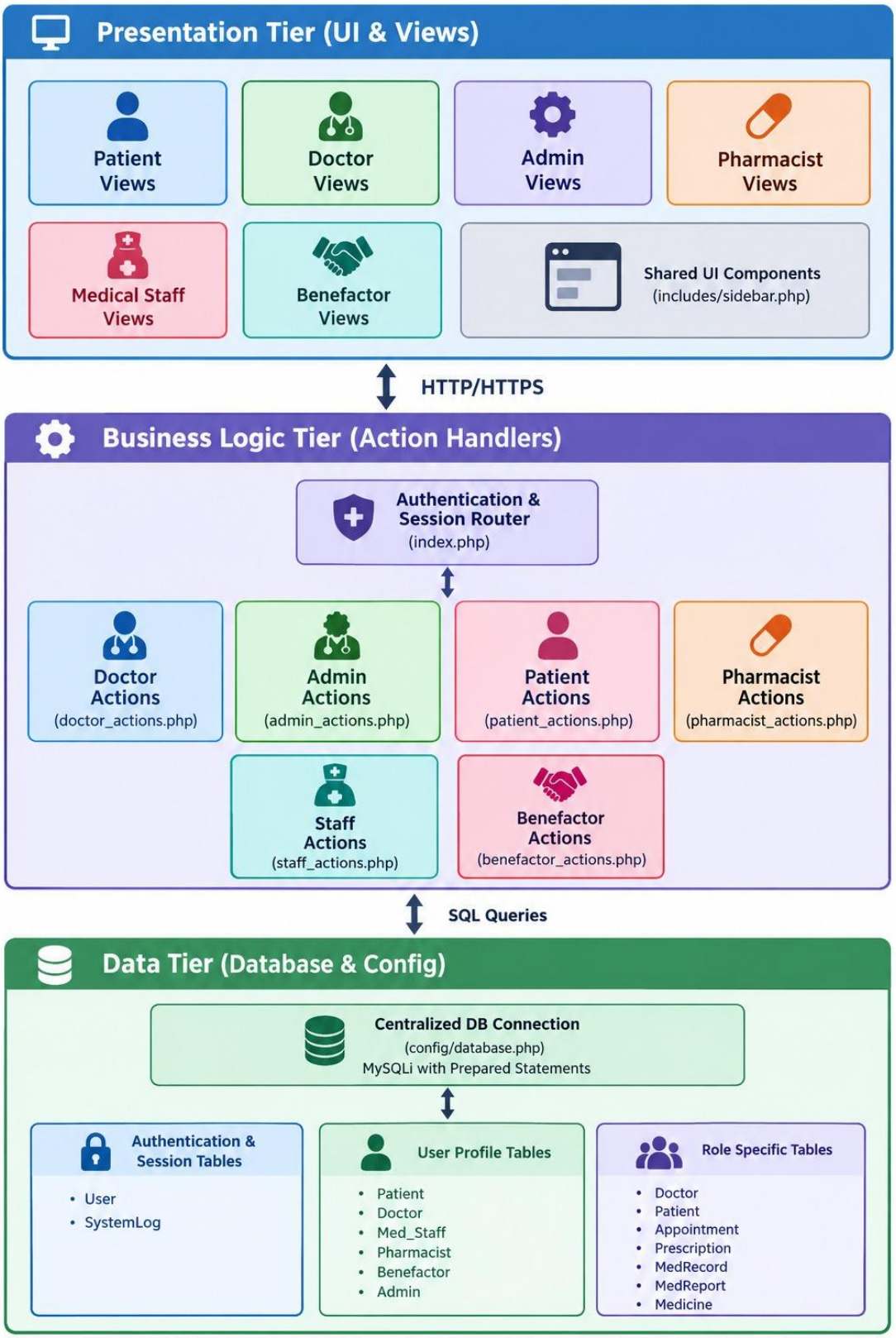
4. Proposed System Architecture

4.1 Architecture Overview

The Cancer Patient Care System follows a Modular (Feature-Based) Architecture. Instead of separating the codebase by technical layers like traditional MVC, the system is organized by business actors and features. Each actor has a dedicated, self-contained module encapsulating their specific UI, business logic, and database interactions.

This architecture is divided into three functional tiers:

- Presentation Tier (Views & Shared UI): Built with HTML5, CSS3, and PHP templating. It consists of role-specific views and shared UI components, and is responsible solely for rendering the interface and capturing user input.
- Business Logic Tier (Action Handlers & Modules): Consists of actor-specific action handlers. These scripts process forms, validate sessions, enforce security checks, and execute business rules, acting as a secure bridge between the UI and the database.
- Data Tier (Shared Configuration & Storage): Comprises the centralized database connection and the MySQL relational database. It manages data persistence using a Class Table Inheritance pattern (a base User table with role-specific extensions) and stores shared static assets.


Presentation Tier (UI & Views)
Patient Views
Doctor Views
Admin Views
Pharmacist Views
Medical Staff Views
Benefactor Views
Shared UI Components (includes/sidebar.php)
HTTP/HTTPS
Business Logic Tier (Action Handlers)
Authentication & Session Router (index.php)
Doctor Actions (doctor_actions.php)
Admin Actions (admin_actions.php)
Patient Actions (patient_actions.php)
Pharmacist Actions (pharmacist_actions.php)
Staff Actions (staff_actions.php)
Benefactor Actions (benefactor_actions.php)
SQL Queries
Data Tier (Database & Config)
Centralized DB Connection (config/database.php)
MySQLi with Prepared Statements
Authentication & Session Tables
User
SystemLog
User Profile Tables
Patient
Doctor
Med_Staff
Pharmacist
Benefactor
Admin
Role Specific Tables
Doctor
Patient
Appointment
Prescription
MedRecord
MedReport
Medicine

4.2 Components and Functionalities

Component	Functionality
Authentication & User Component	Registration, login/logout, role-based access and user account administration.
Patient Component	Profile, appointments, records, prescriptions, meal plans, motivation resources and transport schedules.
Doctor Component	Appointment decisions, patient medical-status updates, prescriptions and treatment monitoring.
Medical Staff Component	Patient Registration, Patient information updates, schedule coordination and medical-report uploads, Manageing patient support requests , Corrdinating with the benefactors & confirming approved assitance.
Pharmacy Component	Stock-quantity updates.
Benefactor Component	Benefactor profiles, donation information and donation history.
Admin & Reporting Component	User administration, system monitoring, configuration and reports.

4.3 Component Interactions

Users access the system through role-specific web interfaces. Requests are sent to PHP modules, where authentication, authorization, validation and business rules are applied. PHP reads from or writes to MySQL as required. The resulting information is returned to the appropriate dashboard. Shared data such as appointments, patient records and prescriptions connects the Patient, Doctor and Medical Staff components, while administrative reports aggregate information from relevant modules.

5. System Design Diagrams

5.1 Use Case Diagram

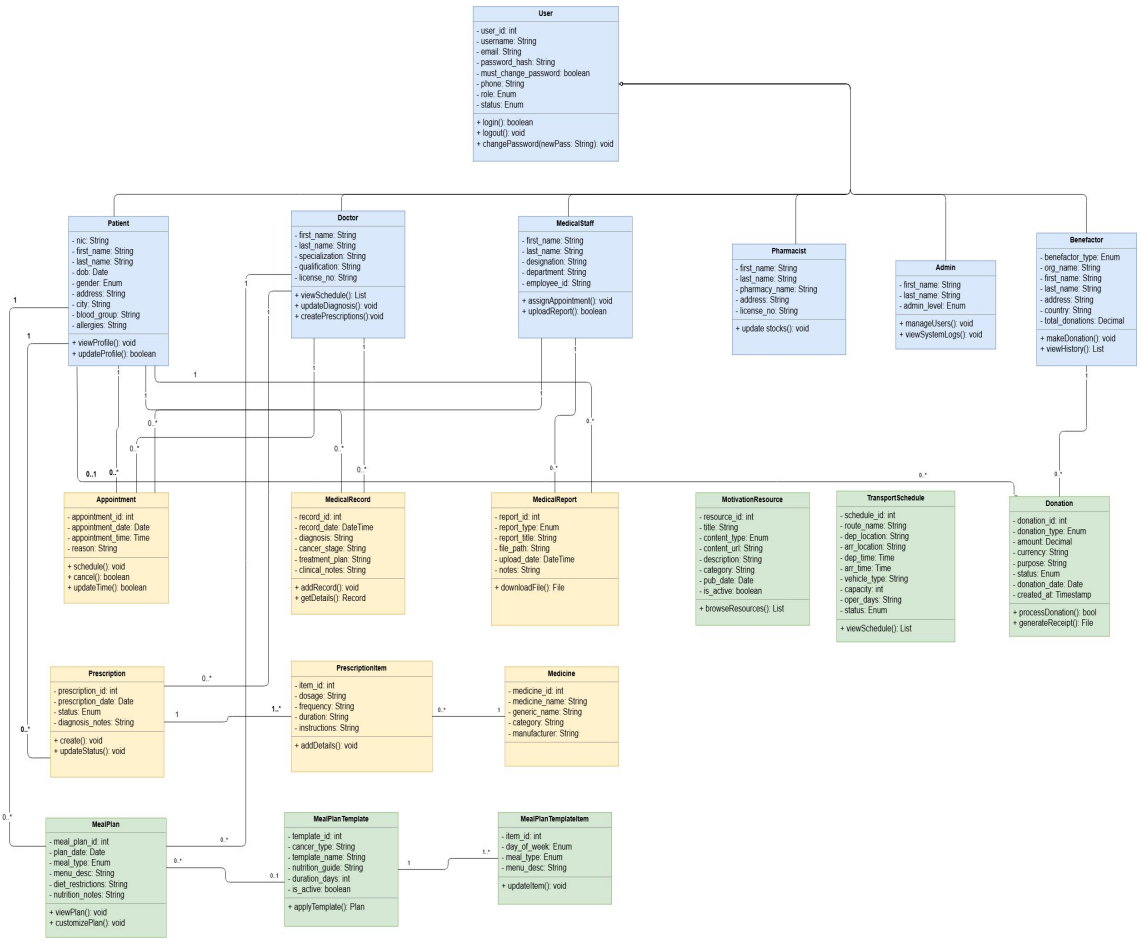
The proposal contains an overall use case diagram and separate actor-specific use case diagrams for Admin, Donor, Patient, Doctor, Medical Staff and Pharmacy. These diagrams show the main interactions between each actor and the system.



https://drive.google.com/file/d/1yxdiLs_xLLOextczcdKIQPaHCwq9Ijub/view?usp=sharing

5.2 Class Diagram (Proposed Interim Design)

The following core classes are derived from the functional requirements and can be refined during detailed design: User, Patient, Doctor, MedicalStaff, Pharmacist, Donor, Admin, Appointment, MedicalRecord, Prescription, Medicine, Donation, MealPlan, MotivationResource and TransportSchedule.

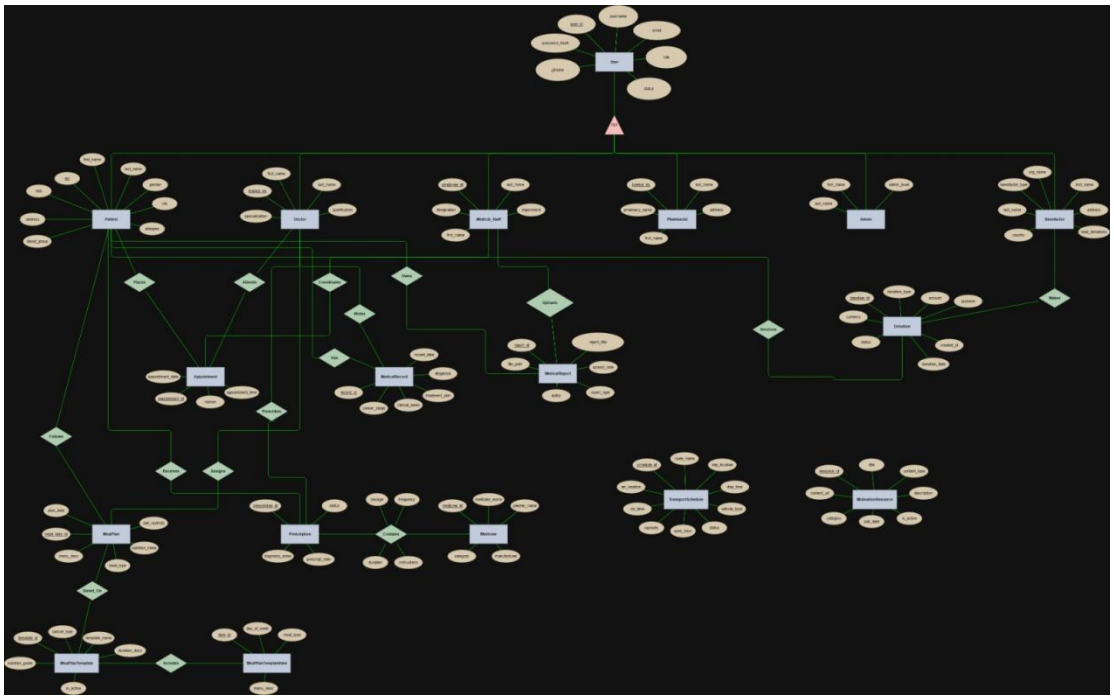


https://drive.google.com/file/d/1KacXbEsHxvnqNnwYbdVh_UgsllVqeXIr/view?usp=sharing

Class	Key Data	Main Relationship
User	user_id, name, email, password, role	Parent concept for role-based users
Patient	patient_id, profile details	Books appointments; has records/prescriptions
Doctor	doctor_id, availability	Reviews appointments; updates records
Appointment	appointment_id, date, status	Links Patient and Doctor
MedicalRecord	record_id, diagnosis, stage, treatment	Belongs to Patient; updated by Doctor/Staff
Prescription	prescription_id, date, details	Created for Patient by Doctor
Medicine	medicine_id, name, quantity	Updated by Pharmacist; viewed by Patient
Benefactor	Benefactor_id, type, date	Linked to Benefactor

5.3 ER Diagram (Proposed Interim Design)

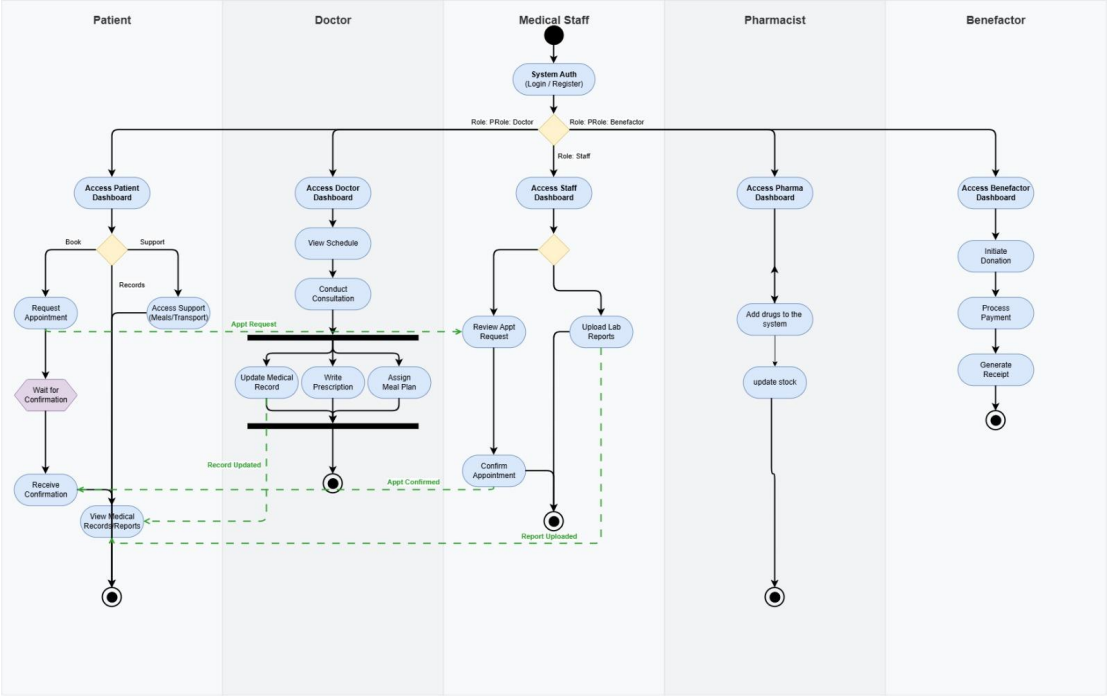
The interim data model centers on USER and PATIENT data, with relational entities for APPOINTMENT, MEDICAL_RECORD, PRESCRIPTION, MEDICINE, BENEFACTOR , MEAL_PLAN, MOTIVATION_RESOURCE and TRANSPORT_SCHEDULE. Primary and foreign keys should be finalized during database implementation.



https://drive.google.com/drive/folders/101BGsb9WDqqT7Eoqn1usncT9z4gZbJAU

5.4 Activity Diagram (Example: Appointment Booking)

1. Patient logs in
2. Selects doctor/date
3. Submits appointment request
4. System validates request
5. Appointment is stored as pending
6. Doctor reviews request
7. Doctor approves, rejects, or reschedules
8. Patient views updated appointment status



6. Current Progress

The following progress statement is an interim estimate based on the approved project timeline and the material already completed in the proposal. The team should replace the percentage with its exact current value before final submission.

Area	Status
Project idea & proposal	Completed
Requirement gathering and analysis	Completed / refined
Functional & quality requirements	Completed / being refined
Use case modelling	Completed at proposal level
Database / ER design	Completed
Class and activity diagrams	Completed
UI design	Completed
Frontend development	Completed
Backend development	Early stage / in progress
Database implementation	Early stage / in progress
Integration and testing	Remaining
Final documentation and presentation	Remaining

Estimated overall completion for interim reporting: approximately 35%. This is a planning estimate inferred from the proposal timeline, not a confirmed measurement of the team's actual implementation progress.

6.1 Remaining Work

- Complete backend development.
- Validate the MySQL database.
- Integrate all modules and role-based access.
- Perform integration, functional and user-acceptance testing.
- Fix defects, evaluate the system, finalize documentation and prepare the final presentation.

6.2 Member Contributions / Responsibilities

Member	Area / Component
W.G.P.V.H. Wickramasinghe	Patient module
M.S.N. Jayasekara	Medical Staff module
K.D. Punsara	Doctor module
P.H.H.R. Rashmika	Pharmacy module
G.R. Milara	Benefactor module
Shared responsibility	Administrator functions, integration, testing, documentation and presentation

7. Technologies

category	Technology
Frontend	HTML, CSS, JavaScript
Backend	PHP
Database	MySQL
Local Server	Apache via XAMPP
Development Tools	Visual Studio Code, GitHub

8. Conclusion

The interim work establishes the problem context, feasibility, requirements and initial system design for the Cancer Patient Care System. The next major focus is to finalize the design models and complete the implementation and integration of the six actor modules while maintaining security, usability and data confidentiality.

Student Name	Index No.	Signature
W.G.P.V.H. Wickramasinghe	24021172	
M.S.N.Jayasekara	24020478	
K.D.Punsara	24020826	
P.H.H.Rashmika	24020852	
G.R.Milara	24020672	